1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

Normalized protein position (bin, 1=N-term, 20=C-term)

Pomphorhynchus laevis

Acanthamoeba castellanii

Cavenderia fasciculata str. SH3

Dictyostelium discoideum

Dictyostelium purpureum str. QSDP1

Entamoeba dispar SAW760

Entamoeba histolytica

Entamoeba invadens IP1

Entamoeba nuttalli P19

Planoprotostelium fungivorum str. Jena

Tieghemostelium lacteum str. TK

Capitella teleta

Dimorphilus gyrociliatus

Helobdella robusta

Owenia fusiformis

Platynereis dumerilii

Babesia bigemina str. Bond

Babesia bovis T2Bo

Babesia ovata str. Miyake

Babesia sp. Xinjiang

Besnoitia besnoiti str. Bb-Ger1

Cryptosporidium andersoni

Cryptosporidium meleagridis str. UKMEL1

Cryptosporidium muris RN66

Cryptosporidium parvum Iowa II

Cryptosporidium ubiquitum

Cyclospora cayetanensis str. CHN_HEN01

Cystoisospora suis str. Wien I

Eimeria acervulina

Eimeria brunetti

Eimeria maxima str. Weybridge

Eimeria mitis

Eimeria praecox

Eimeria tenella str. Houghton

Gregarina niphandrodes

Hammondia hammondi str. H.H.34

Plasmodium berghei

Plasmodium chabaudi

Plasmodium coatneyi

Plasmodium cynomolgi strain B

Plasmodium falciparum

Plasmodium fragile str. multiple

Plasmodium gaboni

Plasmodium gallinaceum

Plasmodium gonderi

Plasmodium inui San Antonio 1

Plasmodium knowlesi

Plasmodium malariae

Plasmodium ovale curtisi

Plasmodium reichenowi

Plasmodium relictum

Plasmodium sp. gorilla clade G2

Plasmodium vinckei petteri str. CR

Plasmodium vivax North Korean

Plasmodium yoelii yoelii str. 17XNL

Theileria equi strain WA

Theileria orientalis strain Shintoku

Theileria parva str. Muguga

Toxoplasma gondii VEG

Aeropyrum pernix K1

Archaeoglobus fulgidus

Candidatus Korarchaeum cryptofilum OPF8

Cenarchaeum symbiosum A

Haloarcula marismortui

Halobacterium salinarum

Haloferax volcanii DS2

Hyperthermus butylicus DSM 5456

Methanobacterium thermoautotrophicum

Methanobrevibacter smithii ATCC 35061

Methanocaldococcus jannaschii

Methanococcus maripaludis S2

Methanopyrus kandleri AV19

Methanosarcina acetivorans

Methanospirillum hungatei JF-1

Methanothermobacter thermautotrophicus str. Delta H

Nanoarchaeum equitans Kin4-M

Natronomonas pharaonis DSM 2160

Nitrosopumilus maritimus SCM1

Pyrobaculum aerophilum str. IM2

Pyrococcus abyssi

Pyrococcus horikoshii OT3

Saccharolobus solfataricus P2

Sulfolobus acidocaldarius

Thermococcus kodakarensis

Thermofilum pendens Hrk 5

Thermoplasma acidophilum

Fragilariopsis cylindrus CCMP1102

Phaeodactylum tricornutum

Pseudo-nitzschia multistriata

Thalassiosira oceanica str. CCMP1005

Thalassiosira pseudonana

Acinetobacter baumannii AYE

Actinobacillus pleuropneumoniae serovar 5b str. L20

Aeromonas hydrophila subsp. hydrophila ATCC 7966

Agrobacterium fabrum str. C58

Aliivibrio fischeri ES114

Anaplasma phagocytophilum str. HZ

Aquifex aeolicus

Azotobacter vinelandii DJ

Bacillus subtilis

Bacteroides thetaiotaomicron VPI-5482

Bartonella henselae str. Houston-1

Bifidobacterium longum NCC2705

Borrelia burgdorferi

Borreliella burgdorferi B31

Bradyrhizobium diazoefficiens USDA 110

Brucella abortus bv. 1 str. 9-941

Buchnera aphidicola str. APS

Burkholderia pseudomallei 1710b

Campylobacter jejuni subsp. jejuni NCTC 11168 = ATCC 700819

Candidatus Koribacter versatilis Ellin345

Caulobacter crescentus

Caulobacter vibrioides CB15

Chlamydia trachomatis D/UW-3/CX

Chlorobaculum tepidum TLS

Chloroflexus aurantiacus J-10-fl

Clostridioides difficile

Coxiella burnetii RSA 493

Cutibacterium acnes KPA171202

Deinococcus radiodurans

Desulfovibrio vulgaris str. Hildenborough

Dictyoglomus turgidum DSM 6724

Enterobacter cloacae subsp. cloacae ATCC 13047

Enterococcus faecalis V583

Escherichia coli K12

Flavobacterium psychrophilum JIP02/86

Francisella tularensis subsp. tularensis SCHU S4 str. Schu S4

Fusobacterium nucleatum subsp. nucleatum ATCC 25586

Gardnerella vaginalis 0288E

Geobacter sulfurreducens PCA

Gloeobacter violaceus PCC 7421

Haemophilus influenzae Rd KW20

Helicobacter pylori 26695

Klebsiella pneumoniae subsp. pneumoniae MGH 78578

Lactobacillus plantarum WCFS1

Lactococcus lactis subsp. lactis Il1403

Leptospira interrogans serovar Lai str. 56601

Leuconostoc mesenteroides subsp. mesenteroides ATCC 8293

Listeria monocytogenes EGD-e

Lysinibacillus sphaericus C3-41

Mesoplasma florum L1

Micrococcus luteus NCTC 2665

Microcystis aeruginosa NIES-843

Moorella thermoacetica ATCC 39073

Mycobacterium tuberculosis

Mycoplasma pneumoniae M129

Myxococcus xanthus DK 1622

Neisseria meningitidis Z2491

Nostoc punctiforme PCC 73102

Paracoccus denitrificans PD1222

Pasteurella multocida subsp. multocida str. Pm70

Porphyromonas gingivalis W83

Prochlorococcus marinus subsp. marinus str. CCMP1375

Proteus mirabilis HI4320

Pseudomonas aeruginosa

Ralstonia solanacearum GMI1000

Rhizobium leguminosarum bv. viciae 3841

Rhodobacter sphaeroides 2.4.1

Rhodopirellula baltica SH 1

Rhodospirillum rubrum ATCC 11170

Rickettsia prowazekii str. Madrid E

Salinibacter ruber DSM 13855

Salmonella typhimurium

Shigella dysenteriae Sd197

Sinorhizobium meliloti 1021

Staphylococcus aureus

Stenotrophomonas maltophilia K279a

Streptococcus pneumoniae

Streptomyces avermitilis

Streptomyces coelicolor

Streptomyces griseus

Synechocystis sp. PCC 6803

Thermanaerovibrio acidaminovorans DSM 6589

Thermodesulfovibrio yellowstonii DSM 11347

Thermosynechococcus elongatus BP-1

Thermotoga maritima

Thermus thermophilus HB8

Treponema pallidum

Ureaplasma parvum serovar 3 str. ATCC 700970

Vibrio cholerae O1 biovar El Tor str. N16961

Wolbachia endosymbiont of Drosophila melanogaster

Xanthomonas campestris pv. campestris str. ATCC 33913

Yersinia pestis biovar Microtus str. 91001

Lingula anatina

Amblyomma americanum

Argiope bruennichi

Centruroides sculpturatus

Dermacentor andersoni

Dermacentor silvarum

Dermatophagoides pteronyssinus

Dinothrombium tinctorium

Galendromus occidentalis

Haemaphysalis longicornis

Hyalomma asiaticum

Ixodes persulcatus

Ixodes scapularis

Leptotrombidium deliense

Limulus polyphemus

Oppia nitens

Ornithodoros turicata

Panonychus citri

Parasteatoda tepidariorum

Rhipicephalus microplus

Rhipicephalus sanguineus

Sarcoptes scabiei

Stegodyphus dumicola

Stegodyphus mimosarum

Tetranychus urticae

Uloborus diversus

Varroa destructor

Varroa jacobsoni

Chlamydomonas reinhardtii

Chlorella variabilis

Coccomyxa subellipsoidea C-169

Micromonas commoda

Ostreococcus lucimarinus

Volvox carteri f. nagariensis

Branchiostoma lanceolatum

Ichthyophthirius multifiliis str. G5

Oxytricha trifallax str. JRB310

Paramecium tetraurelia

Pseudocohnilembus persalinus

Stentor coeruleus

Stylonychia lemnae str. 130c

Tetrahymena thermophila

Acropora millepora

Actinia equina

Actinia tenebrosa

Clytia hemisphaerica

Dendronephthya gigantea

Exaiptasia diaphana

Hydra vulgaris

Hydractinia symbiolongicarpus

Nematostella vectensis

Orbicella faveolata

Pocillopora damicornis

Stylophora pistillata

Thelohanellus kitauei

Folsomia candida

Orchesella cincta

Amphibalanus amphitrite

Cherax quadricarinatus

Daphnia carinata

Daphnia magna

Daphnia pulex

Daphnia pulicaria

Eriocheir sinensis

Eurytemora carolleeae

Homarus americanus

Homarus gammarus

Hyalella azteca

Lepeophtheirus salmonis

Macrobrachium nipponense

Penaeus chinensis

Penaeus japonicus

Penaeus monodon

Penaeus vannamei

Pollicipes pollicipes

Portunus trituberculatus

Procambarus clarkii

Tigriopus californicus

Chroomonas mesostigmatica CCMP1168

Cryptomonas paramecium str. CCAP977/2A

Guillardia theta CCMP2712

Bolinopsis microptera

Mnemiopsis leidyi

Anneissia japonica

Asterias rubens

Lytechinus pictus

Lytechinus variegatus

Patiria miniata

Strongylocentrotus purpuratus

Angomonas deanei

Leishmania donovani str. BPK282A1

Leishmania infantum

Leishmania major

Leishmania panamensis str. MHOM/PA/94/PSC-1

Leptomonas pyrrhocoris str. H10

Leptomonas seymouri str. ATCC 30220

Perkinsela sp. CCAP 1560/4

Phytomonas sp. isolate Hart1

Strigomonas culicis

Trypanosoma brucei

Trypanosoma conorhini

Trypanosoma cruzi marinkellei

Trypanosoma equiperdum

Trypanosoma rangeli SC58

Trypanosoma theileri

Aspergillus fumigatus

Aspergillus nidulans

Candida albicans

Cryptococcus neoformans

Mycosarcoma maydis

Neurospora crassa

Puccinia graminis

Rhizopus delemar

Saccharomyces cerevisiae

Schizosaccharomyces pombe

Zymoseptoria tritici

Emiliania huxleyi

Ptychodera flava

Saccoglossus kowalevskii

Naegleria gruberi

Acromyrmex echinatior

Acyrthosiphon pisum

Adelges cooleyi

Aedes aegypti

Aedes albopictus

Aethina tumida

Agrilus planipennis

Amblyteles armatorius

Amyelois transitella

Anastrepha ludens

Anastrepha obliqua

Ancistrocerus nigricornis

Anopheles albimanus

Anopheles arabiensis

Anopheles atroparvus

Anopheles christyi

Anopheles coluzzii

Anopheles culicifacies

Anopheles darlingi

Anopheles dirus

Anopheles epiroticus

Anopheles farauti

Anopheles funestus

Anopheles gambiae

Anopheles maculatus

Anopheles melas

Anopheles merus

Anopheles minimus

Anopheles quadriannulatus

Anopheles sinensis

Anopheles stephensi

Anoplophora glabripennis

Anthonomus grandis grandis

Aphidius gifuensis

Apis dorsata

Apis florea

Apis mellifera

Athalia rosae

Atta cephalotes

Bactrocera dorsalis

Bactrocera latifrons

Bactrocera neohumeralis

Bactrocera oleae

Bactrocera tryoni

Belgica antarctica

Bemisia tabaci

Bicyclus anynana

Bombus affinis

Bombus huntii

Bombus impatiens

Bombus terrestris

Bombus vancouverensis nearcticus

Bombyx mandarina

Bombyx mori

Bradysia coprophila

Camponotus floridanus

Cataglyphis hispanica

Ceratitis capitata

Chelonus insularis

Chrysoperla carnea

Cimex lectularius

Contarinia nasturtii

Copidosoma floridanum

Coremacera marginata

Cotesia glomerata

Cryptotermes secundus

Ctenocephalides felis

Culex pipiens pallens

Culex quinquefasciatus

Culicoides brevitarsis

Culicoides sonorensis

Cydia pomonella

Cylas formicarius

Daktulosphaira vitifoliae

Danaus plexippus

Dendroctonus ponderosae

Diabrotica virgifera virgifera

Diaphorina citri

Diorhabda carinulata

Diorhabda sublineata

Diuraphis noxia

Drosophila albomicans

Drosophila ananassae

Drosophila arizonae

Drosophila biarmipes

Drosophila bipectinata

Drosophila busckii

Drosophila elegans

Drosophila erecta

Drosophila eugracilis

Drosophila ficusphila

Drosophila grimshawi

Drosophila guanche

Drosophila gunungcola

Drosophila hydei

Drosophila innubila

Drosophila kikkawai

Drosophila mauritiana

Drosophila melanogaster

Drosophila miranda

Drosophila mojavensis

Drosophila montana

Drosophila nasuta

Drosophila navojoa

Drosophila novamexicana

Drosophila obscura

Drosophila persimilis

Drosophila pseudoobscura

Drosophila rhopaloa

Drosophila santomea

Drosophila sechellia

Drosophila serrata

Drosophila simulans

Drosophila subobscura

Drosophila subpulchrella

Drosophila sulfurigaster albostrigata

Drosophila suzukii

Drosophila takahashii

Drosophila teissieri

Drosophila tropicalis

Drosophila virilis

Drosophila willistoni

Drosophila yakuba

Dufourea novaeangliae

Eufriesea mexicana

Glossina austeni

Glossina brevipalpis

Glossina fuscipes

Glossina morsitans morsitans

Glossina pallidipes

Glossina palpalis gambiensis

Glyphotaelius pellucidus

Habropoda laboriosa

Halyomorpha halys

Harpegnathos saltator

Heliconius melpomene

Helicoverpa armigera

Helicoverpa zea

Hermetia illucens

Homalodisca vitripennis

Hylaeus anthracinus

Hylaeus volcanicus

Ichneumon xanthorius

Ischnura elegans

Leguminivora glycinivorella

Leptinotarsa decemlineata

Limnephilus lunatus

Limnephilus marmoratus

Limnephilus rhombicus

Linepithema humile

Lucilia cuprina

Lucilia sericata

Lutzomyia longipalpis

Machimus atricapillus

Macrosteles quadrilineatus

Malaya genurostris

Manduca sexta

Mayetiola destructor

Megachile rotundata

Megaselia scalaris

Melanaphis sacchari

Melitaea cinxia

Microctonus aethiopoides

Microctonus hyperodae

Microplitis demolitor

Microplitis mediator

Monomorium pharaonis

Musca domestica

Myopa tessellatipennis

Nasonia vitripennis

Neodiprion lecontei

Neodiprion pinetum

Nilaparvata lugens

Onthophagus taurus

Ooceraea biroi

Orussus abietinus

Pectinophora gossypiella

Pediculus humanus corporis

Phlebotomus argentipes

Phlebotomus papatasi

Phlebotomus perniciosus

Photinus pyralis

Plodia interpunctella

Pogonomyrmex barbatus

Polistes canadensis

Polistes dominula

Polistes fuscatus

Rhagoletis pomonella

Rhodnius prolixus

Rhopalosiphum maidis

Schistocerca americana

Schistocerca cancellata

Schistocerca gregaria

Schistocerca nitens

Schistocerca piceifrons

Schistocerca serialis cubense

Sipha flava

Sitophilus oryzae

Solenopsis invicta

Spodoptera frugiperda

Stomoxys calcitrans

Teleopsis dalmanni

Tenebrio molitor

Thrips palmi

Topomyia yanbarensis

Toxorhynchites rutilus septentrionalis

Trialeurodes vaporariorum

Tribolium castaneum

Tribolium madens

Trichogramma pretiosum

Uranotaenia lowii

Venturia canescens

Vespa mandarinia

Wyeomyia smithii

Zerene cesonia

Zeugodacus cucurbitae

Zootermopsis nevadensis

Giardia duodenalis ATCC 50581

Giardia intestinalis

Giardia lamblia P15

Kipferlia bialata

Spironucleus salmonicida

Tritrichomonas foetus str. K

Aplysia californica

Biomphalaria glabrata

Crassostrea angulata

Crassostrea virginica

Dreissena polymorpha

Gigantopelta aegis

Haliotis rubra

Haliotis rufescens

Limnoperna fortunei

Liolophura japonica

Lottia gigantea

Magallana gigas

Mercenaria mercenaria

Mizuhopecten yessoensis

Mya arenaria

Mytilus californianus

Octopus bimaculoides

Octopus sinensis

Ostrea edulis

Patella pellucida

Patella vulgata

Pecten maximus

Pomacea canaliculata

Chamberlinius hualienensis

Scutigera coleoptrata

Strigamia maritima

Ascaris suum

Brugia malayi

Caenorhabditis brenneri

Caenorhabditis briggsae

Caenorhabditis elegans

Caenorhabditis japonica

Caenorhabditis remanei

Loa loa

Necator americanus

Onchocerca volvulus

Pristionchus pacificus

Steinernema hermaphroditum

Strongyloides ratti

Trichinella spiralis

Trichuris muris

Gordionus sp. m RMFG-2023

Lineus longissimus

Tubulanus polymorphus

Achlya hypogyna str. ATCC 48635

Aphanomyces astaci

Aphanomyces invadans

Globisporangium iwayamae DAOM BR242034

Globisporangium ultimum DAOM BR144

Hyaloperonospora arabidopsidis

Nothophytophthora sp. Chile5

Peronospora effusa

Phytophthora cactorum str. 10300

Phytophthora infestans

Phytophthora kernoviae

Phytophthora lateralis

Phytophthora megakarya str. zdho120

Phytophthora nicotianae P1569

Phytophthora palmivora var. palmivora str. sbr112.9

Phytophthora parasitica str. IAC_01/95

Phytophthora ramorum

Phytophthora sojae

Plasmopara halstedii

Pythium aphanidermatum

Pythium arrhenomanes

Pythium irregulare

Pythium vexans

Saprolegnia diclina VS20

Saprolegnia parasitica CBS 223.65

Thraustotheca clavata str. ATCC 34112

Perkinsus marinus ATCC 50983 str. PmCV4CB5 2B3 D4

Trichoplax adhaerens

Trichoplax sp. H2

Echinococcus granulosus

Hymenolepis microstoma

Schistosoma haematobium

Schistosoma mansoni

Schmidtea mediterranea

Schmidtea nova

Schmidtea polychroa

Amphimedon queenslandica

Ephydatia muelleri

Halichondria panicea

Sycon ciliatum

Priapulus caudatus

Aureococcus anophagefferens

Blastocystis hominis str. Singapore isolate B

Capsaspora owczarzaki

Ectocarpus siliculosus str. Ec 32

Fonticula alba str. ATCC 38817

Hondaea fermentalgiana

Monosiga brevicollis

Nannochloropsis gaditana CCMP526

Salpingoeca rosetta

Sphaeroforma arctica

Symbiodinium microadriaticum str. CCMP2467

Thecamonas trahens ATCC 50062

Bigelowiella natans

Plasmodiophora brassicae

Reticulomyxa filosa

Chondrus crispus

Cyanidioschyzon merolae

Galdieria sulphuraria

Adineta vaga

Brachionus plicatilis

Rotaria sordida

Hypsibius exemplaris

Paramacrobiotus metropolitanus

Actinidia chinensis

Adiantum capillus-veneris

Aegilops tauschii

Aegilops umbellulata

Amborella trichopoda

Ananas comosus

Arabidopsis halleri

Arabidopsis lyrata

Arabidopsis thaliana

Arabis alpina

Arachis hypogaea

Aristolochia fimbriata

Asparagus officinalis

Avena atlantica CC7277

Avena byzantina PI258586

Avena eriantha BYU132

Avena insularis BYU209

Avena longiglumis CN58138

Avena magna x Avena longiglumis

Avena occidentalis CN25955

Avena sativa Sang

Avena sterilis TN4

Beta vulgaris

Brachypodium distachyon

Brassica juncea

Brassica napus

Brassica oleracea

Brassica rapa R-o-18

Cajanus cajan

Camelina sativa

Cannabis sativa female

Capsicum annuum

Ceratodon purpureus

Ceratopteris richardii

Chara braunii

Chenopodium quinoa

Cinnamomum micranthum f. kanehirae

Citrullus lanatus

Citrus clementina

Closterium sp. NIES-54

Coffea canephora

Corchorus capsularis

Corylus avellana

Corymbia citriodora

Cryptomeria japonica

Cucumis melo

Cucumis sativus

Cynara cardunculus Scolymus

Daucus carota

Digitaria exilis

Dioscorea rotundata

Diphasiastrum complanatum

Echinochloa crus-galli

Eragrostis curvula

Eragrostis tef

Eucalyptus grandis

Eutrema salsugineum

Ficus carica

Fraxinus excelsior

Glycine max

Glycine soja

Gossypium raimondii GPD5lz

Helianthus annuus

Hordeum vulgare ZDM02064

Ipomoea triloba

Juglans regia

Kalanchoe fedtschenkoi

Klebsormidium nitens

Lablab purpureus Natoba

Lactuca sativa Salinas

Lathyrus sativus

Leersia perrieri

Lolium perenne

Lupinus angustifolius

Malus domestica Golden

Manihot esculenta

Marchantia polymorpha

Medicago truncatula

Musa acuminata

Nicotiana attenuata

Nymphaea colorata Beijing-Zhang1983

Olea europaea

Oryza barthii

Oryza brachyantha

Oryza glaberrima

Oryza glumipatula

Oryza longistaminata

Oryza meridionalis

Oryza nivara

Oryza punctata

Oryza rufipogon

Oryza sativa

Panicum hallii HAL2

Papaver somniferum

Phaseolus vulgaris

Physcomitrium patens

Pinus longaeva

Pistacia vera

Pisum sativum JI2822

Populus trichocarpa

Prunus avium

Prunus dulcis

Prunus persica

Quercus lobata

Quercus suber HL8

Rosa chinensis

Saccharum spontaneum

Secale cereale

Selaginella moellendorffii

Sesamum indicum

Setaria italica

Setaria viridis

Solanum lycopersicum

Solanum tuberosum

Sorghum bicolor

Sphagnum fallax

Sphenostylis stenocarpa

Taxus chinensis

Theobroma cacao Matina 1-6

Trifolium pratense

Triticum aestivum Stanley

Triticum dicoccoides

Triticum spelta

Triticum timopheevii

Triticum turgidum

Triticum urartu

Vicia faba

Vigna angularis

Vigna radiata

Vigna unguiculata IT97K-499-35

Vitis vinifera

Zea mays

Convolutriloba macropyga

Hofstenia miamia

Symsagittifera roscoffensis

LCR positional distribution across invertebrate proteomes (purity ≥ 70%, bins 1 & 20 are terminal)

0.00

0.02

0.04

0.06

0.08

0.10

0.12

0.14

Fraction of LCRs in bin